

Module 4: Power Query Editor & Data Transformation

Overview

Data collected from various sources is rarely clean and ready for analysis. It often contains missing values, duplicates, inconsistent formats, incorrect data types, and other quality issues.

Power Query is Power BI's data preparation and transformation engine. It allows analysts to clean, transform, combine, and reshape data before loading it into the data model.

This module covers Power Query fundamentals, data transformation techniques, query management, and best practices for preparing high-quality datasets.

Learning Objectives

After completing this module, you will be able to:

- Understand Power Query Editor
 - Navigate the Power Query interface
 - Perform data cleaning operations
 - Transform data efficiently
 - Handle missing values and duplicates
 - Merge and append queries
 - Pivot and unpivot data
 - Understand Query Folding
 - Use Applied Steps effectively
 - Prepare datasets for analysis
-

Table of Contents

1. Introduction to Power Query
2. Why Data Transformation is Important
3. Power Query Architecture
4. Opening Power Query Editor
5. Power Query Interface
6. Data Types
7. Data Profiling Tools
8. Data Cleaning Techniques
9. Column Transformations
10. Row Transformations
11. Handling Missing Values
12. Handling Duplicates
13. Splitting & Merging Columns

14. Pivot & Unpivot Operations
15. Group By Operations
16. Merge Queries
17. Append Queries
18. Query Folding
19. Applied Steps
20. Best Practices
21. Key Terminologies
22. Practice Exercises
23. Mini Project
24. Module Summary

1 Introduction to Power Query

What is Power Query?

Power Query is a data connection and transformation tool used to:

- Connect to data sources
- Clean data
- Transform data
- Combine datasets
- Prepare data for reporting

It acts as the ETL layer of Power BI.

ETL Process

ETL stands for:

```
graph TD; A[Extract] --> B[Transform]; B --> C[Load];
```

Extract
↓
Transform
↓
Load

Extract

Retrieve data from sources.

Transform

Clean and modify data.

Load

Load transformed data into Power BI.

2 Why Data Transformation is Important

Raw data often contains problems.

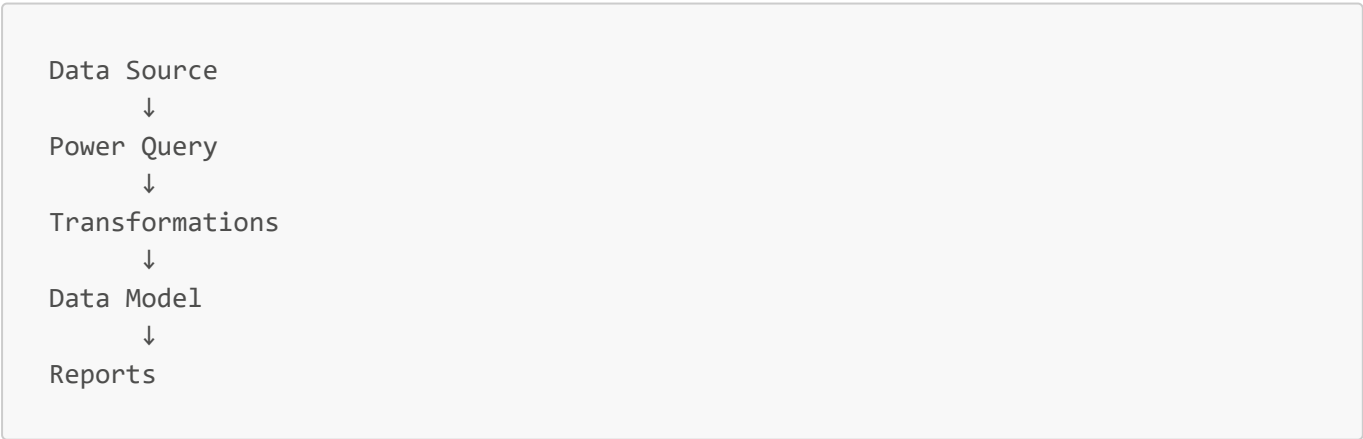
Examples:

Problem	Example
Missing Values	Blank Sales Amount
Duplicates	Same Customer Appears Twice
Wrong Data Types	Date Stored as Text
Inconsistent Formatting	DELHI, Delhi, delhi
Extra Spaces	" Rahul "

Consequences of Poor Data Quality

- Incorrect Reports
- Wrong Business Decisions
- Inaccurate KPIs
- Poor Dashboard Performance

3 Power Query Architecture



Transformation Storage

Power Query does not immediately change source data.

Instead:

- Records transformation steps
- Replays them during refresh

This makes transformations repeatable.

Opening Power Query Editor

Method 1

Home → Transform Data

Method 2

During data import:

Get Data → Transform Data

Power Query Interface

The Power Query Editor contains several important areas.

Ribbon

Contains commands for:

- Transformations
 - Data Management
 - Query Operations
-

Queries Pane

Located on the left side.

Displays:

- Tables
 - Queries
 - Connections
-

Data Preview Area

Displays current data.

Applied Steps Pane

Displays transformation history.

Example:

Source

Changed Type

Removed Columns

Filtered Rows

Formula Bar

Shows transformation formulas.

6

Data Types

Power BI automatically detects data types.

Common Data Types

Data Type	Example
Whole Number	100
Decimal Number	15.75
Currency	₹1000
Text	Delhi
Date	01-Jan-2026
Date/Time	01-Jan-2026 10:30 AM
Boolean	TRUE/FALSE

Why Data Types Matter

Incorrect data types may cause:

- Calculation Errors

- Incorrect Visuals
- Poor Performance

7 Data Profiling Tools

Power Query provides tools to evaluate data quality.

Column Quality

Shows:

- Valid Values
 - Errors
 - Empty Values
-

Column Distribution

Shows frequency of values.

Column Profile

Displays:

- Minimum
 - Maximum
 - Distinct Count
 - Data Distribution
-

Enable Profiling

```
View
↓
Column Quality
Column Distribution
Column Profile
```

8 Data Cleaning Techniques

Data cleaning improves dataset quality.

Common Cleaning Tasks

- Remove Duplicates
- Replace Values
- Trim Spaces
- Change Data Types
- Remove Errors
- Handle Missing Values

Column Transformations

Columns can be modified in various ways.

Rename Column

Right Click → Rename

Remove Column

Home → Remove Columns

Duplicate Column

Creates a copy of an existing column.

Add Custom Column

Example:

Profit = Revenue - Cost

Format Text

Options:

- Uppercase
- Lowercase
- Capitalize Each Word

10 Row Transformations

Rows can also be manipulated.

Filter Rows

Example:

Show only sales greater than ₹10,000.

Sort Rows

Ascending or Descending.

Keep Top Rows

Example:

```
Top 10 Customers
```

Remove Bottom Rows

Useful for eliminating totals.

1 1 Handling Missing Values

Missing values are represented as:

```
null
```

Methods

Remove Rows

Delete rows containing null values.

Replace Values

Replace null with:

0

Unknown

N/A

Fill Down

Copies previous value downward.

Example:

Before
A
null
null

↓

After
A
A
A

12

Handling Duplicates

Duplicates create inaccurate results.

Example

CustomerID
101
101
102

Remove Duplicates

Home → Remove Rows → Remove Duplicates

1 3 Splitting & Merging Columns

Split Column

Example:

Rahul Sharma

Split by space:

First Name	Last Name
Rahul	Sharma

Merge Columns

Combine:

First Name + Last Name

Result:

Rahul Sharma

1 4 Pivot & Unpivot Operations

Pivot

Converts rows into columns.

Before

Month	Sales
Jan	100
Feb	200

After

Jan	Feb
100	200

Unpivot

Converts columns into rows.

Before

Jan	Feb
100	200

After

Month	Sales
Jan	100
Feb	200

Why Unpivot?

Many datasets require unpivoting before analysis.

1 **5** Group By Operations

Similar to SQL GROUP BY.

Example

Sales Data

Region	Sales
North	100
North	200

Group By Result

Region	Total Sales
North	300

Aggregations

- Sum
 - Average
 - Count
 - Min
 - Max
-

Merge Queries

Merge combines tables using a common field.

Similar to SQL JOIN

Example:

Customers Table

CustomerID	Name
1	Rahul

Orders Table

CustomerID	Sales
1	5000

Result

CustomerID	Name	Sales
1	Rahul	5000

Join Types

- Left Outer
- Right Outer

- Full Outer
- Inner Join
- Anti Join

1 7 Append Queries

Append stacks tables vertically.

Table A

Product
Laptop

Table B

Product
Mobile

Result

Product
Laptop
Mobile

Use Cases

- Monthly Sales Files
 - Multiple Branch Data
 - Historical Records
-

1 8 Query Folding

What is Query Folding?

Power Query pushes transformations back to the source database.

Without Query Folding

```
Database
  ↓
Entire Dataset Downloaded
  ↓
Transformations Applied
```

With Query Folding

```
Database
  ↓
Transformation Executed in Database
  ↓
Only Required Data Returned
```

Benefits

- Faster Refresh
 - Reduced Memory Usage
 - Better Performance
-

Applied Steps

Every transformation becomes a step.

Example:

```
Source
Changed Type
Filtered Rows
Removed Columns
Added Custom Column
```

Advantages

- Easy to Modify
 - Easy to Audit
 - Reusable
-

20

Best Practices

Remove Unnecessary Columns

Reduces model size.

Set Correct Data Types

Improves performance.

Use Query Folding

Whenever possible.

Clean Data Before Modeling

Avoid fixing issues later.

Use Meaningful Column Names

Improves readability.

Avoid Loading Unused Tables

Reduces memory consumption.



Key Terminologies

Term	Definition
Power Query	Data transformation engine
ETL	Extract, Transform, Load
Query	Connection to a data source
Transformation	Data modification step
Pivot	Rows to columns
Unpivot	Columns to rows
Merge	Combine tables horizontally
Append	Combine tables vertically
Query Folding	Processing at source system

Term	Definition
Null	Missing value



Practice Exercises

Exercise 1

Import an Excel dataset.

Perform:

- Rename Columns
 - Remove Columns
 - Change Data Types
-

Exercise 2

Identify and remove duplicate records.

Exercise 3

Replace null values.

Exercise 4

Split a Full Name column into:

- First Name
 - Last Name
-

Exercise 5

Merge two datasets using CustomerID.



Mini Project

Customer Sales Data Cleaning

Dataset Contains

- Missing Values
- Duplicate Records
- Incorrect Data Types

- Extra Spaces

Tasks

1. Clean Dataset
2. Remove Duplicates
3. Handle Missing Values
4. Merge Customer Table with Sales Table
5. Create Clean Final Dataset



Module Summary

In this module, you learned:

- ✓ Power Query Fundamentals
- ✓ ETL Process
- ✓ Power Query Interface
- ✓ Data Profiling
- ✓ Data Cleaning Techniques
- ✓ Column & Row Transformations
- ✓ Handling Missing Values
- ✓ Removing Duplicates
- ✓ Pivot & Unpivot
- ✓ Merge Queries
- ✓ Append Queries
- ✓ Query Folding
- ✓ Applied Steps

These skills form the foundation of professional data preparation and are essential before building data models and relationships.



Next Module

Module 5: Data Modeling Fundamentals

Topics Covered:

- Data Modeling
- Relationships

- Cardinality
- Cross Filter Direction
- Primary Keys
- Foreign Keys
- Relationship Best Practices
- Model Optimization